AN19

SILICON DAA LAYOUT GUIDELINES

1. Description

The Si3034/35/36/38/44/46/48 chipsets provide a very high level of integration for modem designs. Integration of the analog front end (AFE) and hybrid, and the removal of the transformer, relays, and opto-couplers reduces the layout effort considerably. Skyworks Solutions' DAA chipset consists of two devices—the DSP-side and the line-side. When designing with the Si3034/35/36/38/44/46/48, there are several layout guidelines that will assist the designer in attaining telecom, safety, and EMC approvals.

This document is divided into six main sections: operational items, analog performance related items, EMC items, safety items, assembly items, and thermal considerations. Operational items are defined as those items that must be followed to ensure functionality of the solution. Analog performance related items are layout recommendations that are pertinent to the analog performance of the solution. EMC items are those items that will affect the emissions/immunity performance of the solution. Safety items are layout issues that could impact safety requirements of a particular modem solution. Assembly items are items that should be noted to assist in assembly of the solution. Thermal considerations are those items that may affect operational performance due to extreme temperatures.

2. Operational Guidelines

Figures 11–14, on pages 11–14, show the typical application circuits for the DAA section. Figure 1 depicts the placement of the chipset, some of the major discrete components, and the RJ11 connector. Note the placement of the Si302x (DSP-side chip) and the Si301x (line-side chip). Aligning these devices so that pins 9–16 of the DSP-side chip face pins 1–8 of the line-side chip will aid in following some of the more specific layout guidelines. Utilizing this placement will also allow the design to closely resemble the Skyworks Solutions' example layout, enabling the designer to directly follow these guidelines.

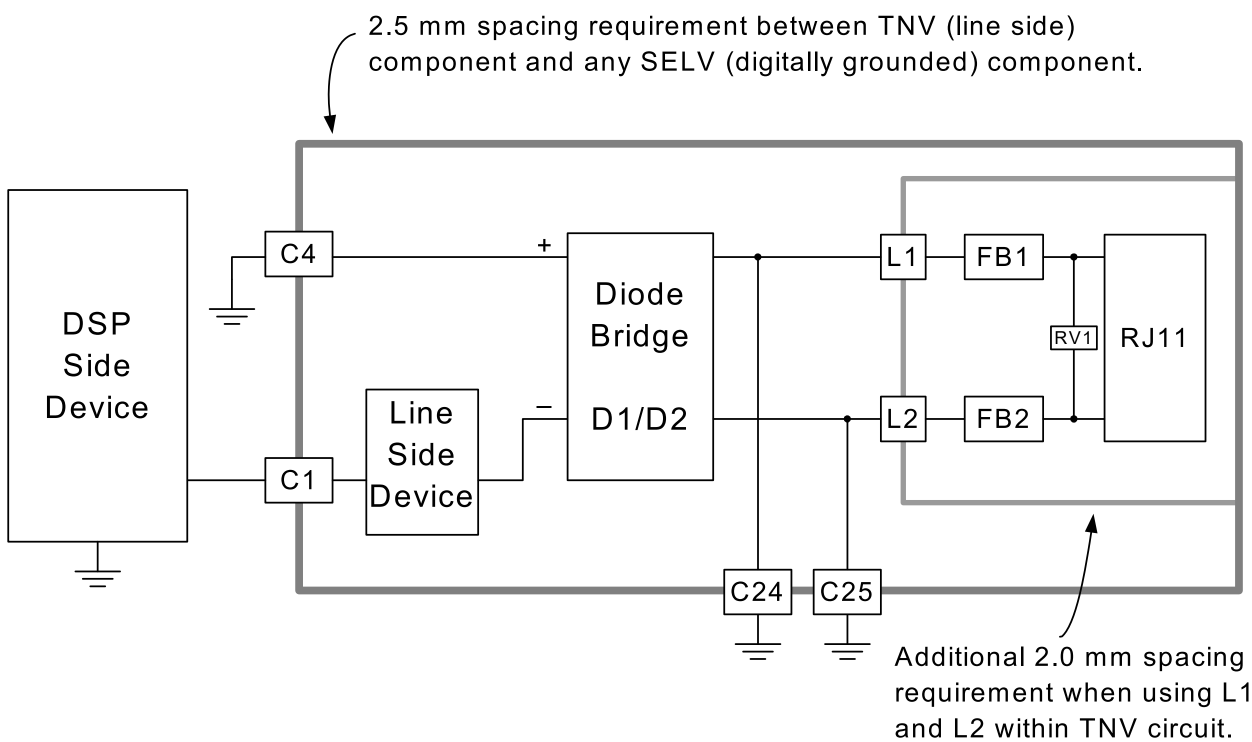


Figure 1. Chipset Diagram

2.1. Si3021/24/25 Layout Requirements

The primary layout considerations for the DSP-side chip (U1) are the placement of the bypass capacitors (C3 and C10) and, if implemented, the filter resistor (R3) for the power supply pins. Capacitors C3 and C10 are the bypass capacitors for V_A and V_D, respectively. A 0.1 µF capacitor provides sufficient bypassing. However, for improved surge immunity the capacitor on V_A should be 0.22 µF and be used in conjunction with the 5.6 V zener diode (Z4). If both V_A and V_D are derived from the same 5 V supply, it is recommended that R3 be used to provide a low-pass filter for V_A. For separate supplies, it is assumed the V_A supply is a low-noise supply. R3 may still be used in this case to provide local filtering of the V_A supply pin. The typical operating current of the V_A supply pin is 1 mA. A 10 Ω resistor with C3 will provide a low pass corner frequency of 725 kHz, while only reducing the supply to V_A by 10 mV. If the power supplied to U1 is 3.3 V, the charge pump internal to U1 must be used to power V_A. In this case, R3 should be removed, and the V_A pin should only be connected to C3.

Figure 2 shows a typical placement of C3 and C10. The hardware designer should focus on minimizing the length of the C3 connection between the V_A and GND pins and the length of the C10 connection between the V_D and GND pins on U1.

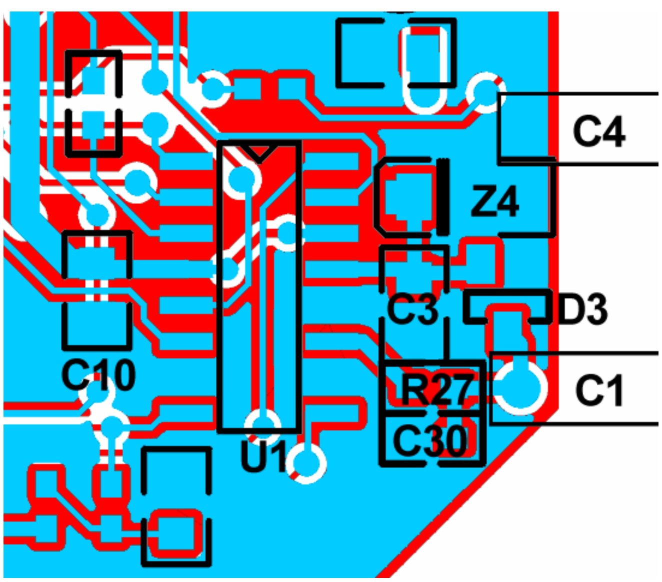


Figure 2. C3 and C10

2.2. Si3024/25 Crystal Layout Requirements

The Si3024/25 incorporates a crystal interface for creating a 24.576 MHz master clock. The Si3024/25 uses the master clock when the device is configured for primary mode. For designs that use a 24.576 MHz crystal as shown in Figure 3, the following layout guidelines should be followed:

- The loop formed by XIN (pin 1), Y1, and XOUT (pin 2) should be minimized and routed on one layer of the PCB.
- The loop formed by Y1, C34, and C35 should be minimized and routed on one layer of the PCB.
- The route from GND (pin 12) to C34 and C35 should be as direct as possible and, if feasible, on one layer of the PCB.

These guidelines will also help minimize issues with emissions. An example layout is shown in Figure 4.

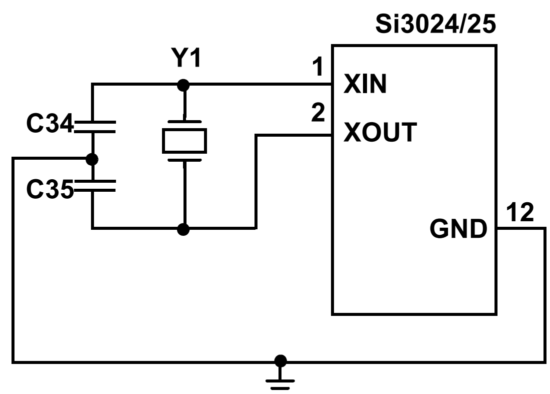


Figure 3. Crystal Circuit

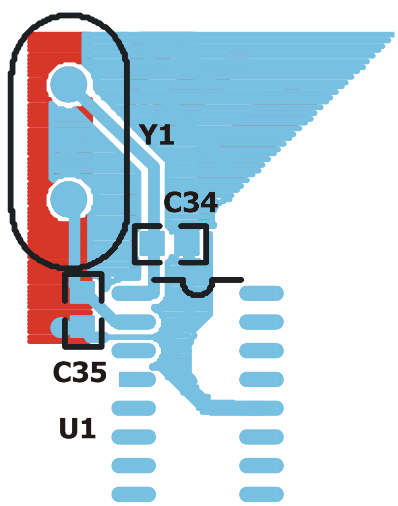


Figure 4. Routing Y1 to U1

2.3. Si3012/14/15 Layout Requirements

For the line-side chip (U2), the primary layout considerations are the placement and routing of C6, C9, C16, D1, D2, and RV2.

Capacitors C6 and C16 provide regulation of the supplies powering U2. The loop formed by C6 to pins 3 and 9 and the loop formed by C16 to pins 3 and 10 should be made as small as possible. Figure 5 shows an example of how to minimize these loops. The trace back to pin 3 is thicker because multiple loops use this path. The thicker trace has a lower impedance, which lessens the effect of multiple currents in that trace.

Capacitor C9, dual diodes D1 and D2, and the MOV RV2 form a loop that should be minimized. C9 performs a low pass filter function on the transmit and receive signals. The inductance in the loop from C9 to the diode bridge (D1 and D2) can affect the ability of C9 to suppress out-of-band energy. See Figure 6 on page 4 for a layout example.

2.4. Si3014/15 Layout Requirements

The placement of capacitors C12 and C13 is important because these components are in high gain loops. Noise in these loops may affect the signals at TIP and RING. The loop formed from pin 15 through C12 to pin 1 (QE2) and the loop formed from pin 16 though C13 to pin 1 should be routed as short as possible. Figure 5 illustrates an example of these loops. Also, the trace from Q4 pin 3 (emitter) should connect directly to the QE2 pin and should not share the same route as the C12 and C13 capacitors to QE2.

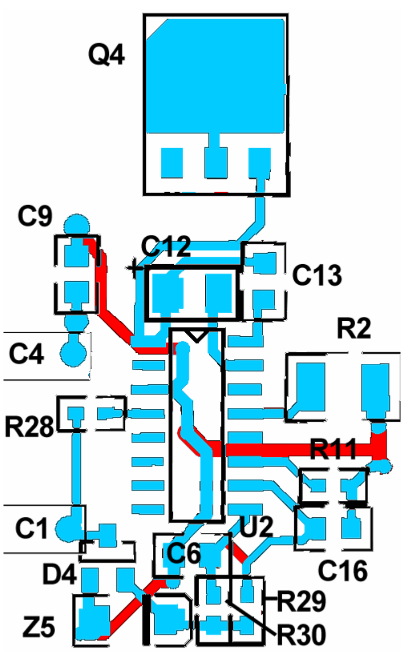


Figure 5. Routing C6, C16, C12, C13, Q4, R2, and R11

3. Analog Performance

The components on the line side of the DAA are composed of a small digital interface section (capacitor barrier interface, converters, and control logic), and the remaining components are used for either analog circuits (hybrid, dc impedance, ac impedance, ringer impedance, etc.) or safety and EMC (surge protector, ferrite beads, EMC capacitors, etc.).

Routing all traces in the DAA section with 15 mil or greater traces when possible will ensure high overall analog performance of the silicon DAA. Furthermore, the DAA section should not use a ground plane for the IGND signal; instead the IGND signal should be routed using a 20 mil trace. Thermal considerations may also be affected by the size of the IGND trace. See "Thermal Considerations" on page 6.

3.1. Si3014/15 Analog Performance

After placing C6, C12, C13, and C16, the designer should focus on placing R11 and routing the trace from Q4 pin 3 to U2 pin 1 using as small a loop as possible to ensure good analog performance. Due to the relatively high current that can flow in the trace from Q4 pin 3 to U2 pin 1, this trace should be routed separately from the trace coming from C12 and C13 to pin 1. This will ensure that the current from Q4 will not affect the sensitive C12 and C13 loops. Figure 5 illustrates an example of the placement and routing of R11, Q4, and U2.

3.2. Si3012 Analog Performance

After placing C6 and C16, the designer should focus on placing R2. When the DAA is off-hook, excess dc current flows through R2. Due to high current, it is important that the IGND trace from R2 be sufficiently wide when other components are connected to the trace as shown in Figure 5.

4. EMC

Several sources conduct or radiate EMC from/to electronic apparatus. Among these are the following:

- Antenna loops formed by ICs and their decoupling capacitors
- PC-board traces carrying driving and driven-chip currents
- Common impedance coupling and crosstalk

To minimize EMC-related problems, all extraneous system noise and the effects of parasitic PC-board trace antennas must be reduced. Employment of an effective system shielding may also be necessary. The following section discusses how to minimize the EMC problems that can occur on the DAA design.

Capacitors C1, C2, C4, and C9 provide the path for the ISOcap™ currents. The typical application schematic recommends that C2 not be installed. If C2 is not installed, the ISOcap current flows in the following manner:

1. From pin 11 of the Si3021/24/25 (U1)
2. Across C1 to pin 4 of the Si3012/14/15 (U2)
3. From pin 4 to pin 3 of U2
4. To C9
5. Across C9 to C4
6. And finally, across C4 to pin 12 of U1

Because the ISOcap signal operates around 2 MHz, it is important to keep the path of the signal as small as possible (see Figure 6).

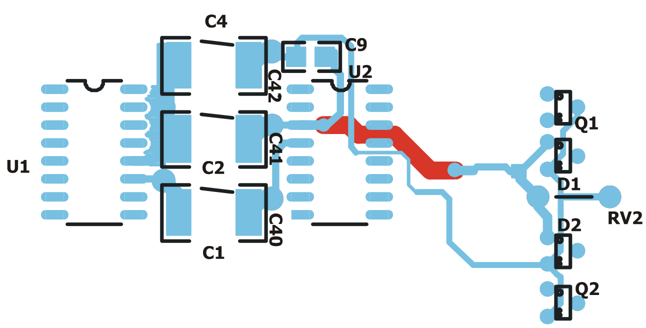


Figure 6. ISOcap™ Interface and Diode Bridge

Short, direct routes using thick 20 mil minimum traces, should be used to connect the ISOcap capacitors to their respective pins. The GND side of these caps should be connected directly to the GND pin (pin 12) on the Si3021/24. The IGND side of C2 should connect directly to the IGND pin of the Si3012/14/15. It is acceptable to use a long trace to connect C4 to the Q1 pin 3 node, as long as there is an IGND trace that follows the C4 to Q1 trace.

For those designs that exhibit emission problems related to the ISOcap interface, the C30 capacitor may be installed. C30 will shunt some of the high frequency energy and reduce emissions due to the harmonics of the ISOcap link.

FB1, FB2, RV1, C24, C25, C31, and C32, and if implemented, a fuse should be placed as close as possible to the RJ11 as shown in Figure 7. It is important for the routing from the RJ11 connector through the ferrite beads FB1 and FB2 to be well matched. The routing to C24 and C25 should also be well matched. The distance from the TIP and RING connections on the RJ11 through the EMC capacitors C24 and C25 to chassis ground should be kept as short as possible. If possible, the routing through the ringer network to the line-side device pin 5 and pin 6 should also be well matched as shown in Figure 8.

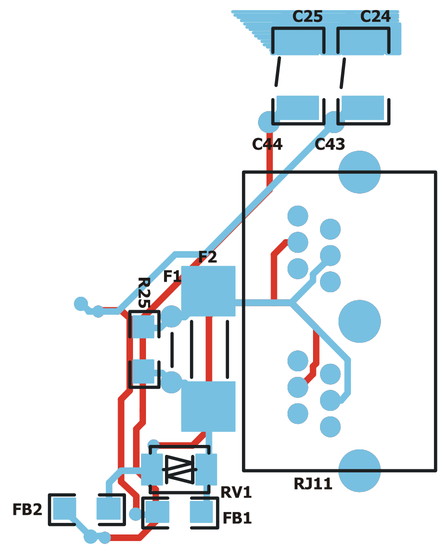


Figure 7. RJ11

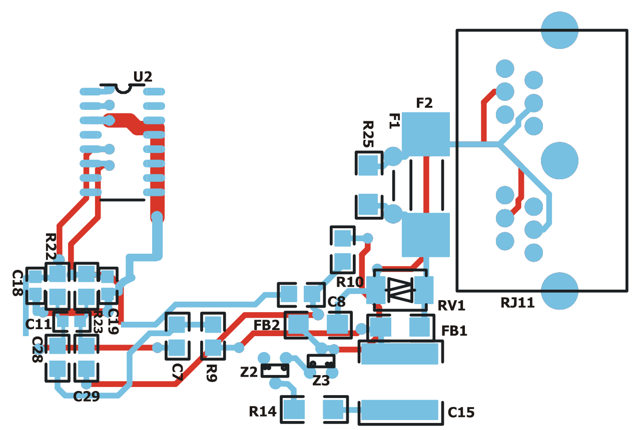


Figure 8. Ringer Network

Routing all the connections from RJ11 to FB1, FB2, RV1, C24, C25, C31, C32, and F1 using a 20 mil trace will improve the EMC performance of the solution.

Good general design practices should be followed to improve the EMC performance of the solution. This includes laying out the digital ground plane as small as possible and rounding off the corners. Placing series resistors on the clock signals near their source and ensuring that the traces from oscillators or crystals are made as short as possible will contribute to the overall emissions performance of the solution.

When using the Skyworks Solutions DAAs in unearthed systems, the designer should consider making some modifications to prevent common mode 50/60 Hz signals from entering the signal path. In an unearthed system, a large 50/60 Hz signal can be present between IGND (line-side pin 3) and DGND (digital-side pin 12). This signal is often as large as 50 V_{RMS}.

The primary consideration is to use well-matched capacitors which bridge the telephone network voltage (TNV) to the safety extra-low voltage (SELV). C1 should match C4, and C24 should match C25. Matching should be within 10%. To the extent that C1 and C24 does not equal C4 and C25, there will be a common mode to differential conversion of the 50/60 Hz signal. Typical common mode rejection of the 50/60 Hz signal is better than 95 dB.

A secondary consideration is capacitive coupling from DGND (SELV) to nodes on the line-side (TNV) circuitry. The most prevalent system design where capacitive coupling occurs is in mini PCI and MDC designs. However, this coupling always exists and can have a noticeable effect on analog performance when SELV is physically close to TNV.

For safety considerations, Skyworks Solutions recommends 3 mm spacing between TNV and SELV (international minimum is 2.5 mm). However, there are three nodes on the line side which require special consideration to insure robust analog performance in unearthed systems:

- RX pin on Si3012/14/15
- FILT pin on Si3014/15
- FILT2 pin on Si3014/15

For these pins and the traces that connect to these pins, the following guidelines should be followed:

- Whenever possible, keep at least a 5 mm distance between these nodes and SELV.
- An IGND guard ring can be used on the board level. This IGND guard ring should be inserted between SELV and the nodes listed above. For FR4 board material, the IGND guard can be placed on any layer. Optimal placement would be for the IGND guard ring to be on the same layer as the nodes listed above. This guard ring allows SELV to couple into IGND rather than the nodes listed above.
- For designs (e.g., MDC, mini PCI designs) which have TNV circuitry in close proximity (< 5 mm) to a SELV plane, a Faraday shield can be used to protect against coupling. This coupling occurs through air between the SELV plane and the TNV nodes described above. A small IGND shield can be placed between the TNV nodes and the SELV plane. This plane allows the SELV to couple into IGND rather than the TNV nodes.

5. Safety

The layout of the modem circuitry, in particular the area of the circuit that is exposed to telephone network voltages (TNV), is subject to many safety compliance issues. Skyworks Solutions recommends that all customers consult with their safety expert or consultant on the various regulations that could impact their designs. Application note “AN17: Designing for International Safety Compliance” discusses the details of safety. Contact your Skyworks Solutions representative to obtain a copy of this document. It contains helpful information regarding multiple safety issues. One of the most critical layout issues that relates to safety is to ensure that a minimum 2.5 mm (3 mm preferable) or 100 mil space is provided between safety extra low voltage (SELV) and TNV circuitry. Designs requiring 5 kV isolation between TNV and SELV should use 5 mm minimum spacing.

When L1 and L2 are included in the DAA, the spacing requirements between the TNV circuit and SELV needs to be increased from 2.5 mm to 3.5 mm. This increased spacing requirement compensates for the secondary peaking effects during longitudinal surges resulting from the addition of L1 and L2. Refer to Figure 1.

In addition to the increased spacing requirement between TNV and SELV, there is a second spacing requirement within the TNV circuit. During surge events, the voltage across the L1 and L2 inductors can be sufficient to create arcing to nearby TNV nodes. To prevent arcing, it is recommended the opposing terminals of the L1 and L2 inductors be isolated from each other by 2.0 mm. Thus, the nodes corresponding to RV1, FB1, FB2, RJ11, and one terminal of L1/L2, should be isolated by 2.0 mm from nodes that connect to the other terminal of L1/L2. Refer to Figure 1.

6. Assembly

There are several steps that can be taken in layout to ensure that the assembly process goes smoothly. For example, an assembly-related error is to install the polarized capacitors with the polarity backwards. This can be prevented by stenciling the board with a plus sign on the correct side of C12, C14, and C5 to indicate the proper orientation for the polarized capacitors. Also, indicating pin 1 on the board with a stencil marking improves the chances that the integrated circuits will be installed correctly. Thought should also be given to using several footprints for a given component to allow for multiple vendor choices. Popular components using multiple footprints are C1, C2, C4, C12, C24, C25, C31, C32, F1, and Z1. Taking these basic steps will assist in the assembly process and ease future troubleshooting.

7. Thermal Considerations

When using the Si3034/44 or Si3038/48 chipset in common applications, there are several thermal considerations that should be taken into account. These thermal considerations will ensure that the device is not operating outside of the recommended operating conditions, thus protecting the device from possible degradation.

On small form factor printed circuit board (PCB) designs, the ability of the PCB to dissipate heat becomes a very important design consideration. These small designs will require additional mechanisms to remove the heat from the Si3014/15. For applications with a PCB area of less than approximately 3000 mm^2, thermal design rules should be applied. These design rules can be found in Skyworks Solutions' application note entitled "Thermal Considerations for Applications using the Si3034, Si3038, and Si3044" (AN21).

8. Check List

Tables 1 and 2 one pages 6 and 9 are check lists that the designer can use during layout. The items marked as required should be taken into consideration first.

Table 1. Si3034/35/44 Check List

✓	#	Layout Items	Required
		Operational Items	
	1	Small loop from C3 (and R3) to U1 pin 13 and pin 12	Yes
	2	Small loop from C10 to U1 pin 4 and pin 12	Yes
	3	Small loop from C6 to U2 pin 9 and pin 3	Yes
	4	Small loop from C16 to U2 pin 10 and pin 3	Yes
	5	Small loop from C12 to U2 pin 15 and pin 1	Yes
	6	Small loop from C13 to U2 pin 16 and pin 1	Yes
	22	Copper pad heat sink is at least 0.08 sq. in. for Q4.	Yes
		Analog Performance	
	7	Small loop from R11 to U2 pin 11 and pin 3	Yes
	8	Separate trace from Q4 pin 3 to U2 pin 1 than the trace from C12 and C13 to U2 pin 1	Yes
	9	Minimum of 15 mil traces in DAA section	
	10	Minimum of 20 mil trace for IGND	
	11	No ground plane in DAA section	Yes
	n/a	> 5 mm spacing between SELV and TNV nodes (RX, FILT, FILT2)	Yes
	n/a	IGND guard ring to protect TNV nodes (RX, FILT, FILT2)	Yes

Table 1. Si3034/35/44 Check List (Continued)

✓	#	Layout Items	Required
	n/a	IGND Farady shield to protect TNV nodes (RX, FILT, FILT2)	Yes
		EMC Items	
	12	Small loop formed between U1, U2, C1, C2, and C4.	Yes
	13	Minimum of 20 mil trace from U1 to C1, C2, and C4 and from U2 to C1 and C2	
	14	FB1, FB2, and RV1 placed as close as possible to the RJ11	Yes
	n/a	Routing from TIP and RING of the RJ11 through F1 to the ferrite beads is well matched	Yes
	16	Distance from TIP and RING through EMC capacitors C24 and C25 to Chassis Ground is short	Yes
	17	C9 is located near C4	
	n/a	C4 is located with C2 between U1 and U2	
	18	Minimum of 20 mil trace from RJ11 to FB1, FB2, RV1, C24, C25, C31, C32, and F1	
	19	Routing of TIP and RING signals from the RJ11 through the ringer networks to U2 pin 5 and pin 6 is well matched	
	20	Trace from D1 & D2 to IGND and to C4–C9 node is well matched and forms a small loop.	
	21	DGND return paths for C10 and C3 should be on component side.	
	n/a	Digital Ground plane is made as small as possible	
	n/a	Ground plane has rounded corners	
	n/a	Series resistors on clock signals are placed near source	
		Safety Items	
	23	Additional 2 mm spacing for inductors (if necessary)	
	24	Minimum 2.5 mm (100 mils) between SELV and TNV (3.0 mm or higher recommended, see "Safety" section)	Yes
	n/a	Space for fire enclosure	
		Assembly Items	
	26	Polarity for C5 is negative side connects with U2 pin 14	
	27	Polarity for C12 is negative side connects with U2 pin 15	
	n/a	Polarity for C14 is negative side connects with U2 pin 3 (IGND)	
	n/a	Pin 1 marking for U1 and U2	

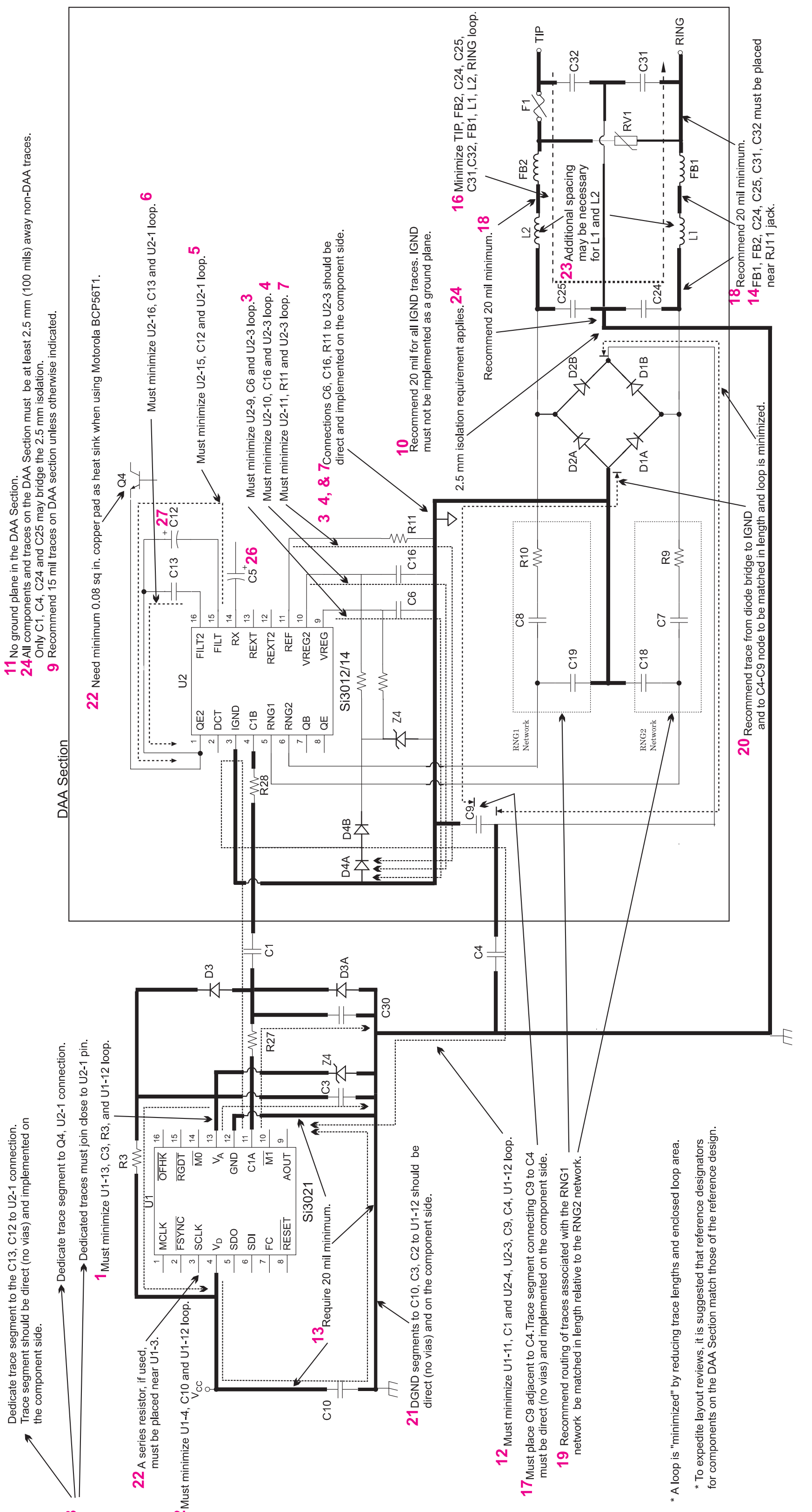


Figure 9. Si3034/35/44 Layout Guidelines

Table 2. Si3036/38/48 Check List

✓	#	Layout Items	Required
		Operational Items	
	1	Small loop from C3 to U1 pin 13 and pin 12	Yes
	2	Small loop from C10 to U1 pin 4 and pin 12	Yes
	3	Small loop from C6 to U2 pin 9 and pin 3	Yes
	4	Small loop from C16 to U2 pin 10 and pin 3	Yes
	5	Small loop from C12 to U2 pin 15 and pin 1	Yes
	6	Small loop from C13 to U2 pin 16 and pin 1	Yes
	22	Copper pad heat sink is at least 0.08 sq. in. for Q4.	Yes
		Analog Performance	
	7	Small loop from R11 to U2 pin 11 and pin 3	Yes
	8	Separate trace from Q4 pin 3 to U2 pin 1 than the trace from C12 and C13 to U2 pin 1	Yes
	9	Minimum of 15 mil traces in DAA section	
	10	Minimum of 20 mil trace for IGND	
	11	No ground plane in DAA section	Yes
	n/a	> 5 mm spacing between SELV and TNV nodes (RX, FILT, FILT2)	Yes
	n/a	IGND guard ring to protect TNV nodes (RX, FILT, FILT2)	Yes
	n/a	IGND Farady shield to protect TNV nodes (RX, FILT, FILT2)	Yes
		EMC Items	
	12	Small loop formed between U1, U2, C1, C2, and C4.	Yes
	13	Minimum of 20 mil trace from U1 to C1, C2, and C4 and from U2 to C1 and C2	
	14	FB1, FB2, and RV1 placed as close as possible to the RJ11	Yes
	n/a	Routing from TIP and RING of the RJ11 trough F1 to the ferrite beads is well matched	Yes
	16	Distance from TIP and RING through EMC capacitors C24 and C25 to Chassis Ground is short	Yes
	17	C9 is located near C4	
	n/a	C4 is located with C2 between U1 and U2	
	18	Minimum of 20 mil trace from RJ11 to FB1, FB2, RV1, C24, C25, C31, C32, and F1	
	19	Routing of TIP and RING signals from the RJ11 through the ringer networks to U2 pin 5 and pin 6 is well matched	
	20	Trace from D1 & D2 to IGND and to C4–C9 node is well matched and forms a small loop.	
	21	DGND return paths for C10 and C3 should be on component side.	
	28	Traces from oscillators or crystals are made as short and direct as possible	
	n/a	Digital Ground plane is made as small as possible	
	n/a	Ground plane has rounded corners	
	n/a	Series resistors on clock signals are placed near source	
		Safety Items	
	23	Additional 2 mm spacing for inductors (if necessary)	
	24	Minimum 2.5 mm (100 mils) between SELV and TNV (3.0 mm or higher recommended, see "Safety" section)	Yes
	n/a	Space for fire enclosure	
		Assembly Items	
	26	Polarity for C5 is negative side connects with U2 pin 14	
	27	Polarity for C12 is negative side connects with U2 pin 15	
	n/a	Polarity for C14 is negative side connects with U2 pin 3 (IGND)	
	n/a	Pin 1 marking for U1 and U2	

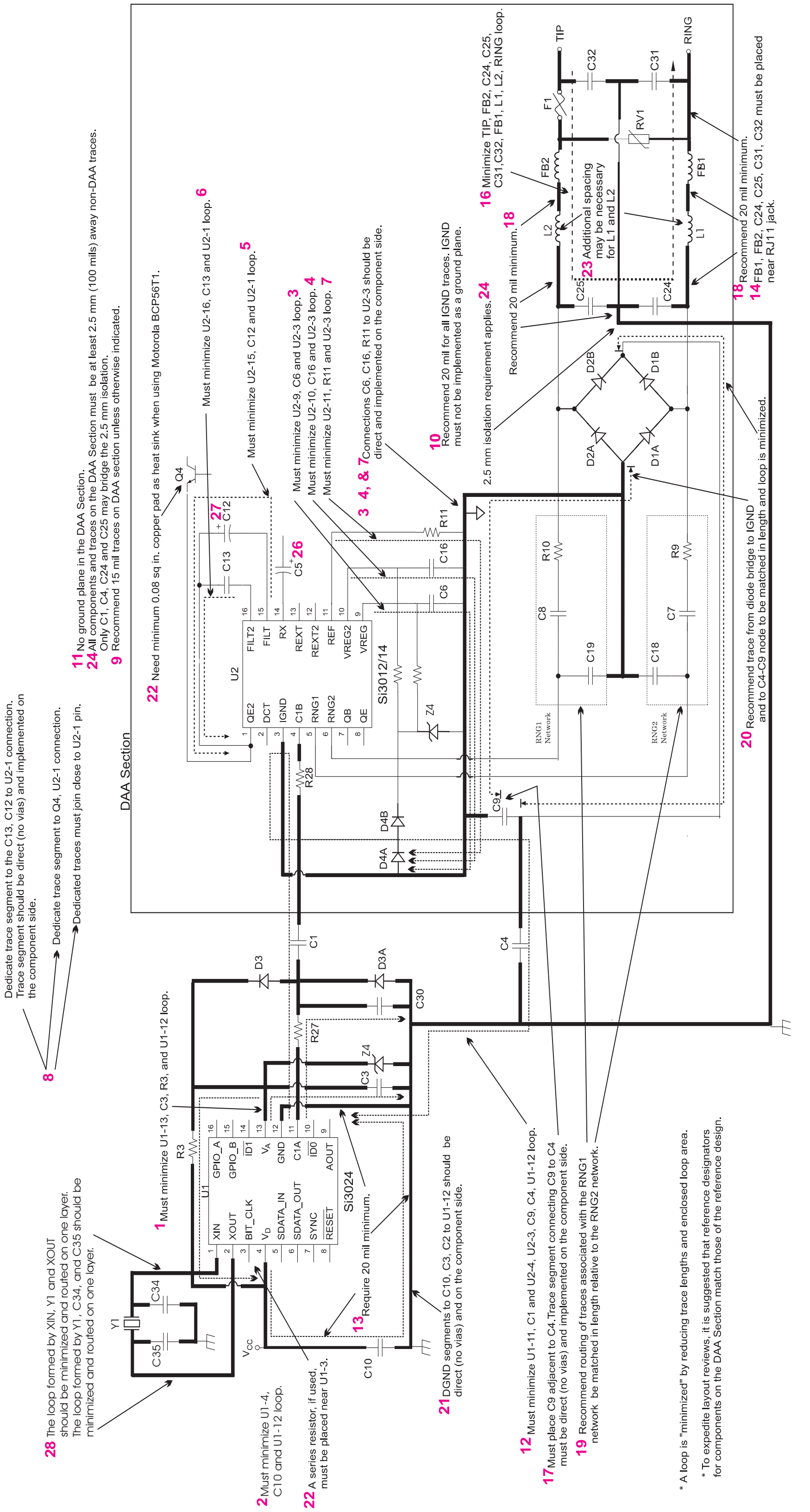


Figure 10. Si3036/38/48 Layout Guidelines

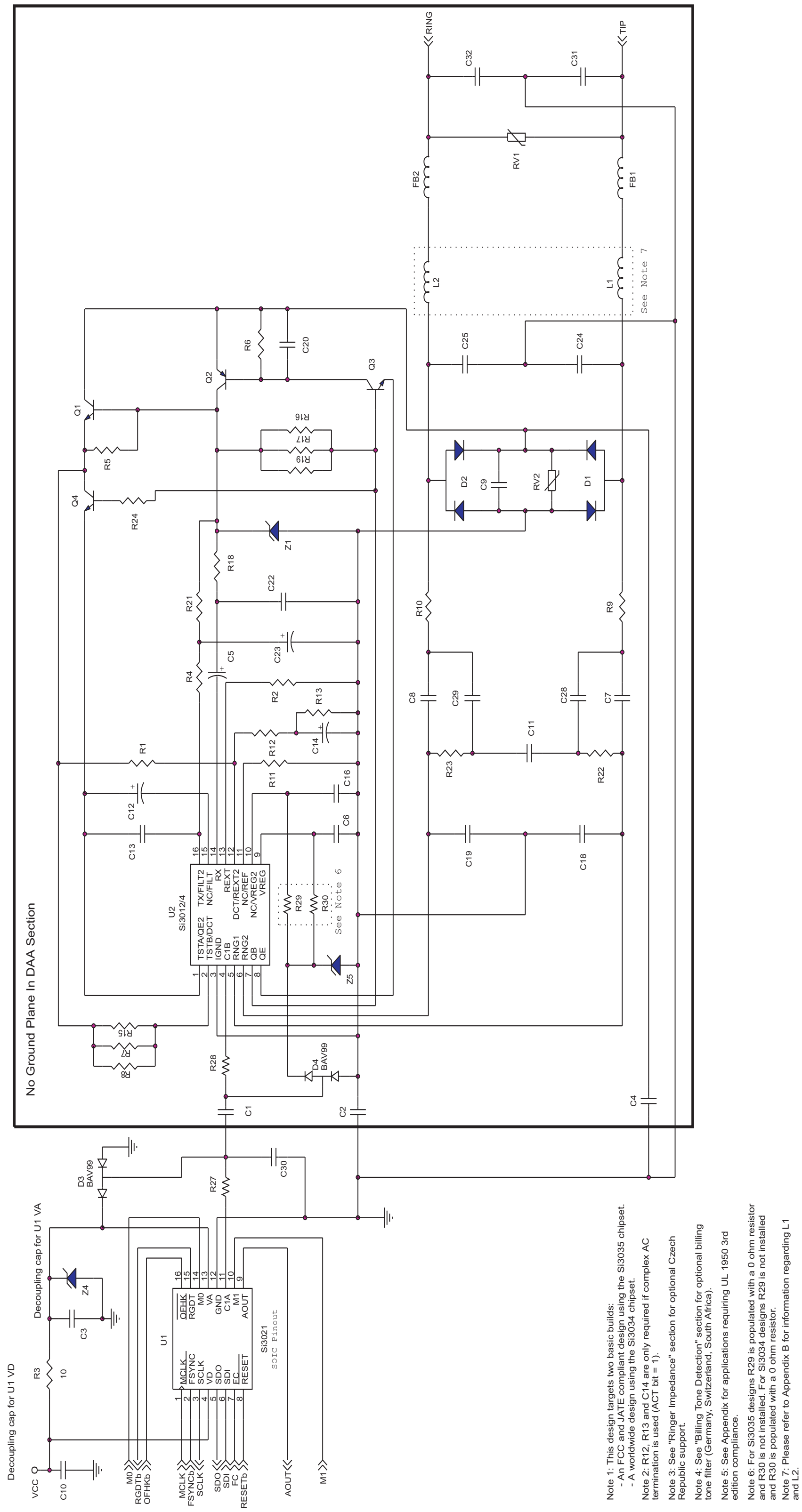
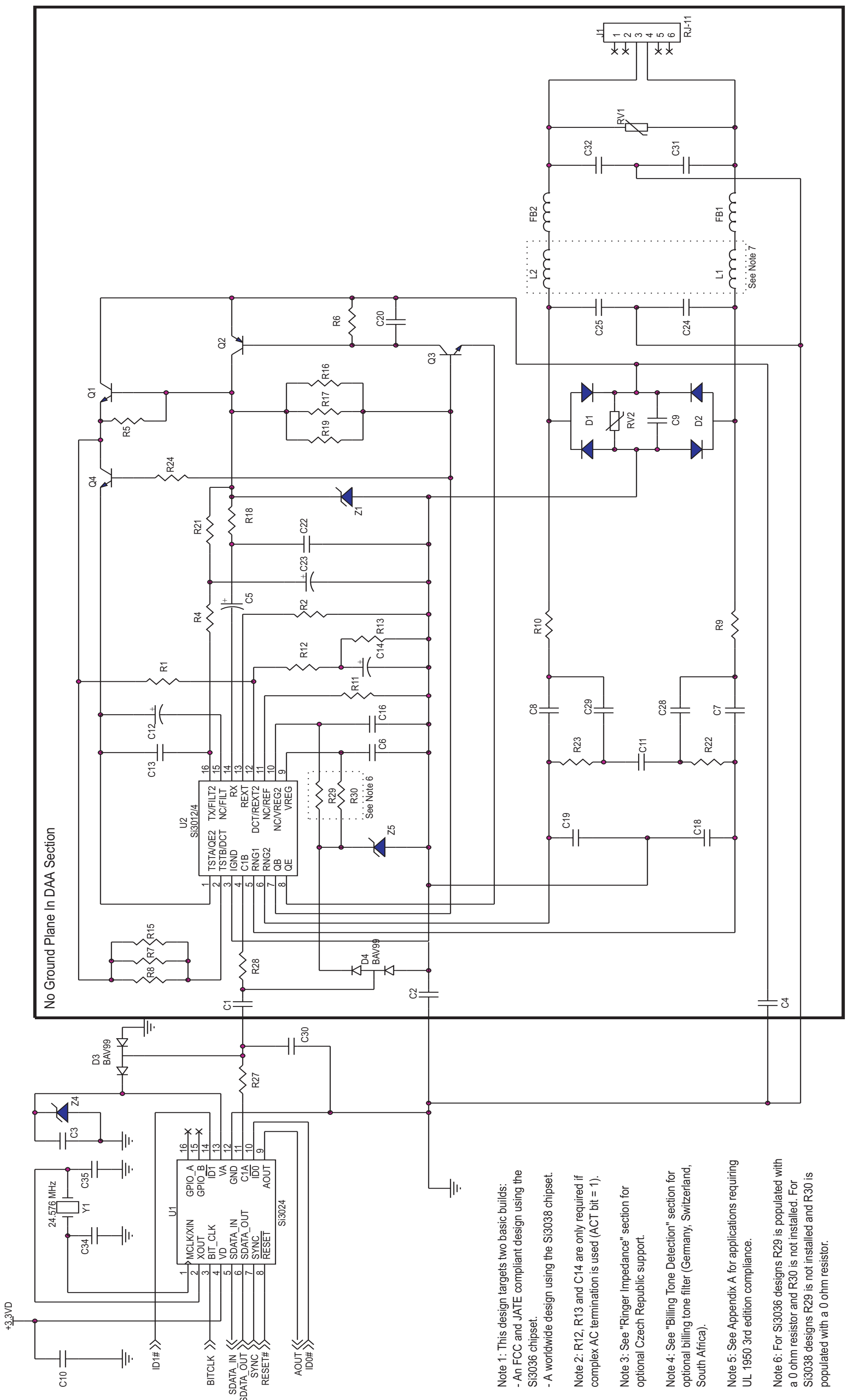


Figure 11. Si3034/35 Typical Application Diagram



Note 1: This design targets two basic builds:
- An FCC and JATE compliant design using the Si3036 chipset.
- A worldwide design using the Si3038 chipset.

Note 2: R12, R13 and C14 are only required if complex AC termination is used (ACT bit = 1).

Note 3: See "Ringer Impedance" section for optional Czech Republic support.

Note 4: See "Billing Tone Detection" section for optional billing tone filter (Germany, Switzerland, South Africa).

Note 5: See Appendix A for applications requiring UL 1950 3rd edition compliance.

Note 6: For Si3036 designs R29 is populated with a 0 ohm resistor and R30 is not installed. For Si3038 designs R29 is not installed and R30 is populated with a 0 ohm resistor.

Note 7: Refer to Appendix B for information regarding L1 and L2.

Figure 12. Si3036/38 Typical Application Diagram

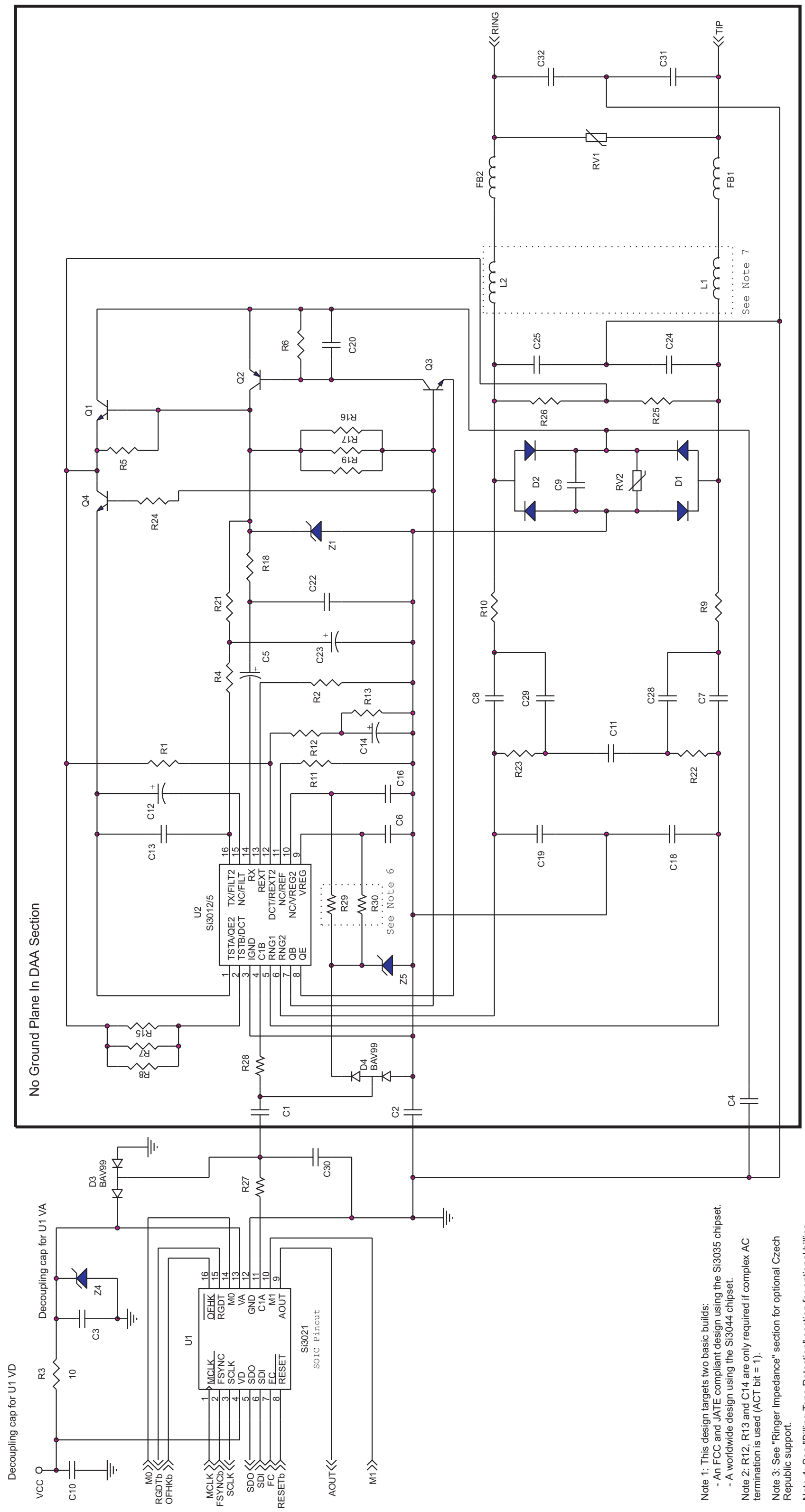


Figure 13. Si3044/35 Typical Application Diagram

Note 1: This design targets two basic builds:
- An FCC and JATE compliant design using the Si3035 chipset.
- A worldwide design using the Si3044 chipset.

Note 2: R12, R13 and C14 are only required if complex AC termination is used (ACT bit = 1).

Note 3: See "Ringer Impedance" section for optional Czech Republic support.

Note 4: See "Billing Tone Detection" section for optional billing tone filter (Germany, Switzerland, South Africa).

Note 5: See Appendix for applications requiring UL 1950 3rd edition compliance.

Note 6: For Si3035 designs R29 is populated with a 0 ohm resistor and R30 is not installed. For Si3044 designs R29 is not installed and R30 is populated with a 0 ohm resistor.

Note 7: Please refer to Appendix B for information regarding the installation of L1 and L2.

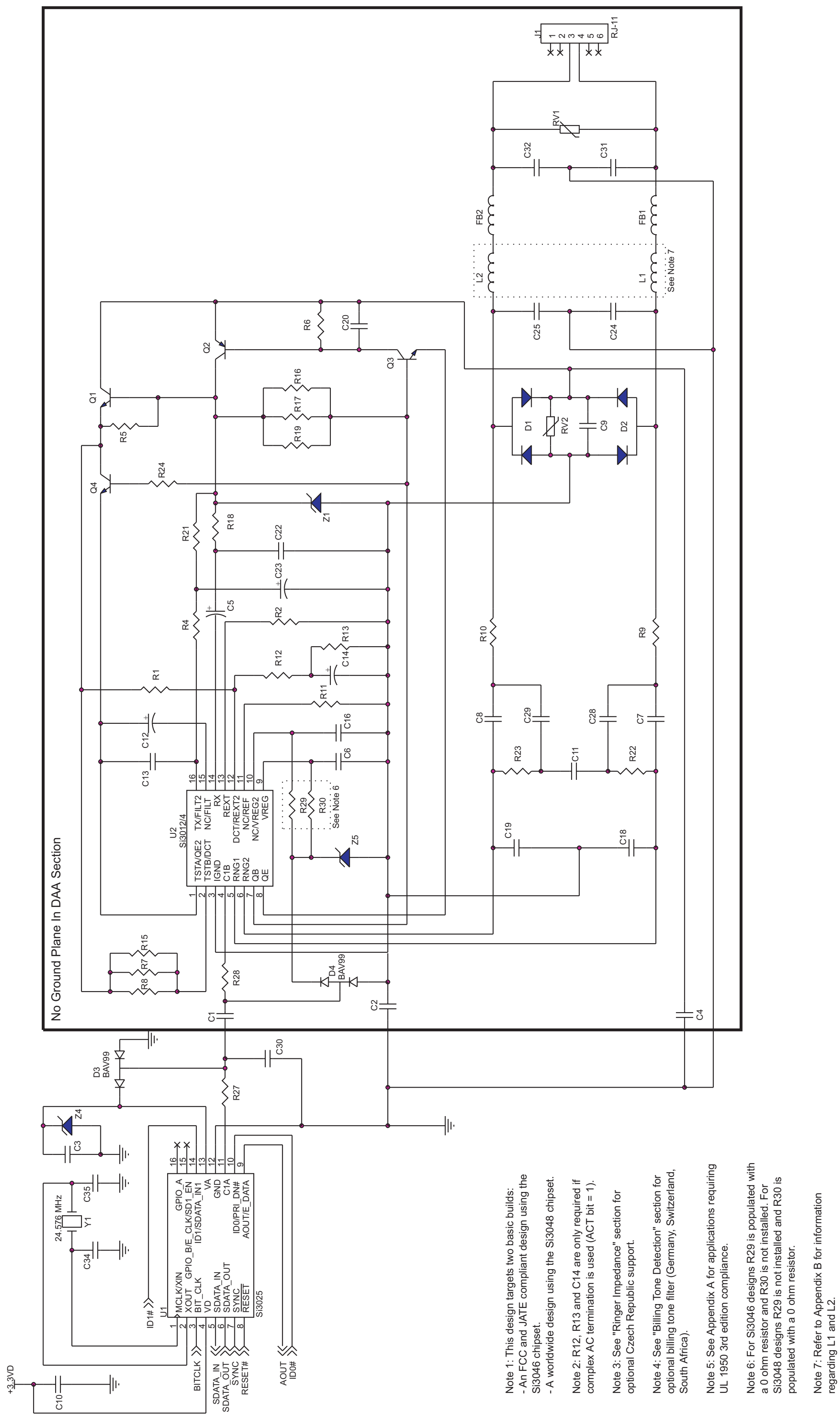


Figure 14. Si3046/48 Typical Application Diagram

NOTES:

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